

6406533952667. ✓ I have written answers on the answer sheets

6406533952668. ✖ Not applicable

LLM

Section Id :	64065382586
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	14
Number of Questions to be attempted :	14
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653181250
Question Shuffling Allowed :	No

Question Number : 117 Question Id : 6406531172694 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533952669. ✓ YES

6406533952670. ✖ NO

Question Number : 118 Question Id : 6406531172695 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

General Note:

- If you find the given information is insufficient in any of the Numerical Answer Type (NAT) questions, enter -1 as the answer.
- In all your calculations, take at least two digits. For example, if the intermediate result is 1.860875, then take it as 1.86 to the next step.

Options :

6406533952671. ✓ Instructions has been mentioned above.

6406533952672. ✗ This Instructions is just for a reference & not for an evaluation.

Sub-Section Number :

2

Sub-Section Id :

640653181251

Question Shuffling Allowed :

Yes

Question Number : 119 Question Id : 6406531172696 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider following assertion and reason pair:

Assertion: A small vocabulary is desirable, as it reduces the size of the embedding matrix.

Reason: Decreasing the size of the vocabulary increases the number of tokens in the input.

Options :

6406533952673. ✗ Assertion and Reason are both true and Reason is a correct explanation of Assertion.

6406533952674. ✗ Assertion and Reason are both true and Reason is not a correct explanation of Assertion.

6406533952675. ✗ Assertion is true but Reason is false.

6406533952676. ✓ Assertion is false but Reason is true.

Question Number : 120 Question Id : 6406531172698 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider following statement and mark if it is true or false:

Repeating examples in the pre-training datasets are completely harmless for downstream performance.

Options :

6406533952680. ✗ TRUE

6406533952681. ✓ FALSE

Sub-Section Number :

3

Sub-Section Id :

640653181252

Question Shuffling Allowed :

Yes

for even 175 B param.
model → even 100 repetition
is harmful

Question Number : 121 Question Id : 6406531172697 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

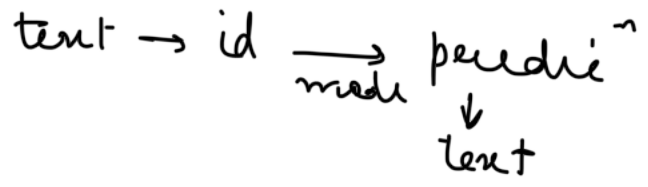
What is the order of the language modeling pipeline?

Options :

6406533952677. ✖ First, the model, which handles text and returns raw predictions. The tokenizer then makes sense of these predictions and converts them back to text when needed.

6406533952678. ✖ First, the tokenizer, which handles text and returns IDs. The model handles these IDs and outputs a prediction, which can be some text.

6406533952679. ✔ The tokenizer handles text and returns IDs. The model handles these IDs and outputs a prediction. The tokenizer can then be used once again to convert these predictions back to some text.



Sub-Section Number :

4

Sub-Section Id :

640653181253

Question Shuffling Allowed :

Yes

Question Number : 122 Question Id : 6406531172699 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider fine-tuning the T5 model on a sentiment classification task. The T5 model was pre-trained using the C4 dataset. Suppose, in the original dataset, the positive samples are denoted by integer 2, neutral samples are denoted by integer 1 and the negative samples are denoted by integer 0. Choose the necessary modifications that one needs to perform to use the T5 model for fine-tuning:

Options :

$+$ $\rightarrow$ 2

0 $\rightarrow$ 1

$-$ $\rightarrow$ 0

α

6406533952682. ✖ Map the label 0 to "negative" and 1 to "positive" in the dataset

6406533952683. ✔ Map the label 0 to "negative", 1 to "neutral" and 2 to "positive" in the dataset

6406533952684. ✔ Map the label 0 to "0", 1 to "1" and 2 to "2" in the dataset

6406533952685. ✖ Update the vocabulary of T5 with the modified labels

6406533952686. ✖ Add at least one linear layer on top of the decoder for final classification

6406533952687. ✖ None of these.

Question Number : 123 Question Id : 6406531172700 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

The strikethrough words in the passage given below denote the words to be dropped from the original sentence.

"These models can't read ~~your mind~~. If outputs are too long, ~~ask for brief replies~~. If outputs are too simple, ask for expert-level writing. If you dislike the format, ~~demonstrate the format~~ you'd like to see. The ~~less the model has to guess~~ at what you want, the more likely you'll get it."

Which of the following represents the correct input sequence, with sentinel tokens, to the baseline model that uses pre-training denoising objectives? [z] represents the end of the sentinel token in a sentence. The characters inside the square brackets are the sentinel tokens.

Options :

6406533952688. ✖ [a] your mind [b] ask for brief replies [c] demonstrate the format [d] less the model has to guess [e] 
6406533952689. ✔ [a] your mind [b] ask for brief replies [c] demonstrate the format [d] less the model has to guess 
6406533952690. ✔ [v] your mind [w] ask for brief replies [x] demonstrate the format [y] less the model has to guess 
6406533952691. ✖ [v] your mind [w] ask for brief replies [x] demonstrate the format [y] less the model has to guess [z]
6406533952692. ✖ [a] your mind [b] demonstrate the format [c] ask for brief replies [d] less the model has to guess 
6406533952693. ✖ None of these

Question Number : 124 Question Id : 6406531172701 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose the T5 base model has to be fine tuned with gradual unfreezing. First layer of the encoder takes the input in appropriate format. Gradual unfreezing is divided in 12 (i.e. no. of layers) phases.

Choose the correct statements:

Options :

6406533952694. ✔ In the first phase, weights of only the last layer of encoder and decoder (both) are unfrozen. 
6406533952695. ✖ In the first phase, weights of only the last layer of decoder only are unfrozen.
6406533952696. ✖ In the first phase, weights of only the last layer of encoder only are unfrozen.
6406533952697. ✔ In the second phase, weights of only the second last layer of encoder and decoder (both) are unfrozen. 
6406533952698. ✖ In the second phase, weights of the last two layers of encoder and decoder (both) are unfrozen.
6406533952699. ✖ In the first phase, weights of only the first layer of encoder and decoder (both) are unfrozen.

Question Number : 125 Question Id : 6406531172702 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following models uses a form of denoising objective for pre-training?

Options :

6406533952700. ✓ BERT

6406533952701. ✗ GPT 3

6406533952702. ✓ BART

6406533952703. ✓ T5

→ Causal LM

Question Number : 126 Question Id : 6406531172703 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the C4 pipeline. Which of the following will NOT pass through it as it is?

Options : C → *eq. cleaning — 5 things to check*
el aprendizaje automatico es genial.

(Above sentence is Spanish translation of "Machine learning is great")

6406533952704. ✓

Following function prints each key-value pair from a dictionary:

```
# printing using For Loop
for key, value in input_dict.items():
    print(f"{key}: {value}")
```

6406533952705. ✓

6406533952706. ✓ *Lorem ipsum dolor sit amet... } ⇒ unnecessary*

Please enable JavaScript to use our site.

Home

Products

Shipping

Contact

FAQ

6406533952707. ✓ ...

had not eq. add

Cricket is a very popular sport in India. Cricket is a very popular sport in India.

6406533952708. ✓ *duplicate*

US tariffs will have negative economic consequences globally, some economists say.

6406533952709. ✗

✓✓

6406533952710. ✖ None of these.

Question Number : 127 Question Id : 6406531172704 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are correct about GPT-3 model with 175B parameters:

Options :

6406533952711. ✖ It is an encoder-decoder only model.

6406533952712. ✔ It uses absolute positional encoding.

6406533952713. ✔ It uses a composite dataset for pre-training.

6406533952714. ✖ None of these.

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

5

640653181254

Yes

^
✓
✓
✓
L4 + wiki +
book +
web text

Question Number : 128 Question Id : 6406531172705 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

Consider a vocabulary $\mathcal{V}$,

$\mathcal{V} = ([\text{start}], \overset{1}{\text{deep}}, \overset{2}{\text{hot}}, \overset{3}{\text{is}}, \overset{4}{\text{learning}}, \overset{5}{\text{research}}, \overset{6}{\text{topic}}, \overset{7}{\text{very}}, \overset{8}{\text{end}}]$.

Assume that the BART model is being trained using a denoising objective.

Suppose the original sentence is “[start] ²deep ³learning ⁴is ⁵very ⁶hot ⁷research ⁸topic [end]” and the corrupted sentence is “[start] deep [mask] is [mask] very hot [mask] topic [end]”

The prediction probabilities outputted by the model are given below.

	1	2	3	4	5	6	7	8	9
1	0.1	0.17	0.11	0.12	0.08	0.17	0.15	0.22	0.28
2	0.07	0.19	0.09	0.12	0.18	0.02	0.03	0.01	0.24
3	0.13	0.21	0.18	0.17	0.09	0.21	0.04	0.16	0.04
4	0.16	0.12	0.07	0.06	0.15	0.12	0.2	0.	0.18
5	0.11	0.15	0.17	0.15	0.07	0.12	0.24	0.02	0.19
6	0.12	0.05	0.02	0.07	0.07	0.15	0.15	0.25	0.03
7	0.04	0.04	0.12	0.06	0.09	0.06	0.05	0.03	0.19
8	0.02	0.05	0.07	0.18	0.02	0.22	0.03	0.24	0.13

The zeroth row of the matrix is the output probability distribution by the model given the special [Start] token as input. Following the Greedy Search decoding strategy, the subsequent rows give the conditional probability distribution conditioned over the previous tokens. What is the loss value?

0.17 $\Rightarrow$ -1.77
0.18 $\rightarrow$ -1.71
0.17 $\rightarrow$ -1.77
0 $\rightarrow$ ∞
0.17 $\rightarrow$ -1.77
0.15 $\rightarrow$ -1.89
0.05 $\rightarrow$ -2.99
0.13 $\rightarrow$ -2.04
640653181255

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.90 to 2.0
Sub-Section Number :
Sub-Section Id :
Question Shuffling Allowed :

Yes

Question Number : 129 **Question Id :** 6406531172706 **Question Type :** SA
Correct Marks : 3

Question Label : Short Answer Question

Suppose you are working on prefix language modeling. The sequence length is 64 and the first 3 tokens represent the task specific prefix. How many non-infinity elements are there in the mask for computing attention scores?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

2115

Sub-Section Number : 7
Sub-Section Id : 640653181256
Question Shuffling Allowed :

No

Question Id : 6406531172707 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix
Question Numbers : (130 to 136)
Question Label : Comprehension

Consider the following dictionary with the number of word occurrences in a corpus:

```
wo = { "taught" : 2,  
       "laughter" : 1,  
       "drought" : 4,  
       "tough" : 5 }
```



Handwritten list of tokens and their counts:

- t ✓
- a ✓
- u ✓
- g ✓
- h ✓
- l ✓
- e ✓
- r ✓
- d ✓
- a ✓
- { /w } ✓

Note: Append </w> to each word at the end.

You will be working with BPE algorithm, answer the given subquestions in that context:

Sub questions

Question Number : 130 Question Id : 6406531172708 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

How many tokens are there in the initial vocabulary?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Handwritten answer: 11

Question Number : 131 Question Id : 6406531172709 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following pairs has the highest frequency before any merge?

Options :

6406533952718. ✓ ('u', 'g')

6406533952719. ✗ ('a', 'u')

6406533952720. ✗ ('g', 'h')

6406533952721. ✗ ('h', 't')

6406533952722. ✗ ('w', 'e')

6406533952723. ✗ None of these.

Handwritten calculation:

$$2 + 1 + 4 + 5 = 12$$

Question Number : 132 Question Id : 6406531172710 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

What is the frequency of the pair merged in the very first merge?

Response Type : Numeric

Evaluation Required For SA : Yes

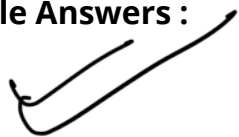
Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12



Question Number : 133 Question Id : 6406531172711 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

What is the minimum number of merges, after which there is a token with exactly 3 English letters?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

ugh
ugh
① ug
② h

Question Number : 134 Question Id : 6406531172712 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Which pair will be merged in the third merge? Say the pair is ('ab','c'), then enter **"abc"** (without quotes and white spaces). If there is a tie between two or more candidates, pick the one that occurs first in the original vocabulary.

NOTE: Enter the exact answer without any extra space in the beginning or at the end.

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

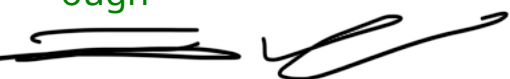
Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

ough



Question Number : 135 Question Id : 6406531172713 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Assume that the process is stopped after 4 merges. How many tokens from the initial vocabulary

will have their frequency revised/reduced to 0?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 136 Question Id : 6406531172714 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Assuming that the algorithm was stopped after 4 merges only, how will the word “trought</w>” be tokenized with this vocabulary? Please note that </w> is already appended to the word.

Options :

~~6406533952728.~~ ✓ t, r, ough, t</w>

6406533952729. ✗ tr, o,ugh, t</w>

6406533952730. ✗ t, r, ou, gh, t</w>

6406533952731. ✗ t, r, ough, t, </w>

6406533952732. ✗ tr, ough, t, </w>

6406533952733. ✗ None of these

t r ough t</w>

i-NLP

Section Id :

64065382587

Section Number :

8

Section type :

Online

Mandatory or Optional :

Mandatory

Number of Questions :

19

Number of Questions to be attempted :

19